MATH 250: Introduction to Statistics

Fall 2025 Syllabus

Instructor Information

Professor: Dr. Emily Chen
Email: e.chen@university.edu
Office: Mathematics Building, Room 312
Office Hours: Tuesday/Thursday 2:00-4:00 PM

Course Information

Meeting Times: MWF 10:00-10:50 AM
Location: Science Hall, Room 204
Credits: 3 semester hours

Course Description

This course provides an introduction to statistical concepts and methods. Topics include descriptive statistics, probability distributions, hypothesis testing, confidence intervals, correlation, and regression. Emphasis is placed on practical applications and interpretation of results.

Learning Outcomes

Upon successful completion of this course, students will be able to:
1. Calculate and interpret descriptive statistics
2. Apply probability theory to solve real-world problems
3. Conduct hypothesis tests and construct confidence intervals
4. Perform correlation and regression analysis

Required Materials

• **Textbook:** *Statistics for Data Science* by Johnson & Lee (3rd ed.)
• **Calculator:** TI-84 or equivalent
• **Software:** R or Python (free, instructions provided)

Grading Policy

Component	Weight	Description
Homework	20%	Weekly assignments (10 total, lowest dropped)
Quizzes	15%	Short quizzes every two weeks (5 total)

Midterm Exam	25%	Covers chapters 1-6
Final Exam	30%	Comprehensive
Participation	10%	Attendance and in-class activities

Letter Grade Scale:
A: 90-100% | B: 80-89% | C: 70-79% | D: 60-69% | F: Below 60%

Course Schedule (Tentative)

Week	Topics	Assignments
1-2	Descriptive Statistics, Data Visualization	HW 1
3-4	Probability Basics, Distributions	HW 2, Quiz 1
5-6	Normal Distribution, Sampling	HW 3
7-8	Confidence Intervals, Hypothesis Testing	HW 4, Quiz 2, Midterm
9-10	t-tests, ANOVA	HW 5, Quiz 3
11-12	Chi-square Tests, Correlation	HW 6
13-14	Linear Regression, Multiple Regression	HW 7, Quiz 4
15	Review and Practice	Quiz 5
16	Final Exam	Final Exam

Course Policies

Attendance: Regular attendance is expected. More than 3 absences may affect your grade.
Late Work: Late homework is accepted within 48 hours with a 20% penalty.
Academic Integrity: All work must be your own. Violations will result in disciplinary action.

Accessibility Statement

Students with disabilities who need accommodations should contact the Office of Disability Services and provide documentation. I am committed to ensuring all students have equal access to course materials and activities.